Mathura — Greater Noida

AI-Based Stock Price Trend Prediction System (AINESP)

Submitted in partial fulfillment of requirements
for the degree of

B.Tech. (Computer Science & Engineering)

By

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GLA/CEA/FR/14/2025-26

GLA University, Mathura

Department of CEA

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This is to certify that the Mini Project entitled **"AI-Based Stock Price Trend Prediction System (AINESP)"** is a bonafide work carried out by **Himanshu Goyal (2415000685)** submitted to GLA University, Mathura in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering.

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The Mini Project report entitled **AI-Based Stock Price Trend Prediction System (AINESP)** by **Himanshu Goyal (2415000685)** is approved for the award of Bachelor's of Technology Degree in **Computer Science & Engineering**.

Supervisor

Date:

Place: Mathura-281406

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Date:

Place: Mathura-281406

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Date:

Place: Mathura-281406

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Supervisor

Date:

Place: Mathura-281406

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or work have been included. I have adequately cited and referenced the original source. I also declare that I have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke the penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature

Name of Student
and Roll No

Date: 2026-05-03

Place: Mathura - 281406

Abstract

Stock market analysis is a complex and dynamic process that involves handling large volumes of time-dependent financial data. Stock prices fluctuate continuously due to various influencing factors such as economic conditions, company performance, market sentiment, and global events. Due to this inherent uncertainty and volatility, accurately predicting stock price trends has always been a challenging task. Traditional approaches rely heavily on manual analysis and basic statistical techniques, which often fail to capture complex patterns in large datasets.

This project presents the design and implementation of an **AI-Based Stock Price Trend Prediction System (AINESP)** aimed at analyzing historical stock data and forecasting future trends using advanced machine learning techniques. The system is developed as a full-stack application, integrating a **FastAPI-based backend** with a **Next.js frontend** to provide real-time visualization and user interaction.

The system utilizes historical stock data obtained from financial APIs such as Yahoo Finance. The collected data undergoes preprocessing steps including data cleaning, normalization, and handling missing values to ensure data quality. Feature engineering techniques are applied to generate important technical indicators such as Moving Averages, Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD), which enhance the predictive performance of the models.

To achieve accurate predictions, a hybrid machine learning approach is adopted. The system combines **Long Short-Term Memory (LSTM)** networks and **eXtreme Gradient Boosting (XGBoost)** models. LSTM is used to capture temporal dependencies and sequential patterns in time-series data, while XGBoost efficiently analyzes feature-based relationships. The outputs of both models are combined to improve prediction accuracy and reliability.

The results are presented through an interactive dashboard that visualizes historical data along with predicted trends using graphical charts. Performance is evaluated using metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and directional accuracy. The system demonstrates strong capability in identifying overall trends, although it has limitations in predicting sudden market fluctuations caused by external events.

This project provides practical exposure to machine learning, data engineering, and full-stack development, while highlighting the potential and limitations of AI in financial forecasting. It serves as an educational tool for understanding stock market analysis and predictive modeling.

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List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
LSTM	Long Short-Term Memory
XGBoost	eXtreme Gradient Boosting
API	Application Programming Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
OHLCV	Open, High, Low, Close, Volume
SMA	Simple Moving Average
EMA	Exponential Moving Average
RSI	Relative Strength Index
MACD	Moving Average Convergence Divergence
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
EDA	Exploratory Data Analysis
ETL	Extract, Transform, Load
UI	User Interface
UX	User Experience
NLP	Natural Language Processing
DFD	Data Flow Diagram

Chapter 1

Introduction

This project presents an **AI-Based Stock Price Trend Prediction System (AINESP)** that uses machine learning techniques to analyze historical stock data and predict future price trends. The system integrates a FastAPI backend with a Next.js frontend to provide real-time visualization and intelligent insights for users.

1.1 Description

The system fetches stock data from financial APIs and processes it using data preprocessing and feature engineering techniques. It applies machine learning models such as **LSTM** and **XGBoost** to generate predictions. The results are displayed through an interactive dashboard.

Figure 1.1: Main dashboard showing stock price history and AI prediction

The dashboard provides graphical representation of historical and predicted prices, making it easier for users to understand trends.

1.2 Problem Formulation

Stock market data is highly volatile and influenced by multiple factors such as economic conditions, company performance, and global events. Traditional analysis methods are not efficient for handling such complex data. Therefore, there is a need for an automated system that can:

- Process large volumes of financial data
- Identify hidden patterns
- Predict future trends accurately

1.3 Motivation

Manual analysis of stock data is time-consuming and prone to human errors. Traditional statistical models fail to capture non-linear patterns in financial data. Machine learning provides a better solution by learning from historical data and improving prediction accuracy.

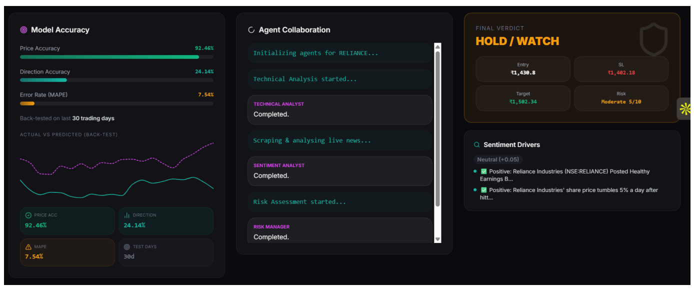

Figure 1.2: Model accuracy showing prediction performance metrics

1.4 Proposed Solution

The proposed system uses a hybrid machine learning approach combining:

- **LSTM** for time-series prediction
- **XGBoost** for feature-based analysis

A multi-agent system is implemented where different agents perform technical analysis, sentiment analysis, and risk assessment to generate a final prediction.

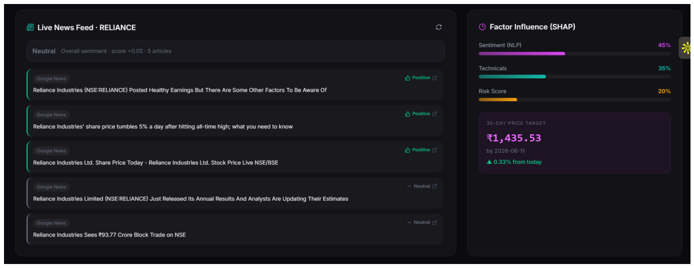

Figure 1.3: Multi-agent system performing analysis tasks

The system provides users with insights such as predicted price, risk level, and final decision (Buy/Hold/Sell).

1.5 Scope of the Project

The project focuses on:

- Historical stock data analysis
- Machine learning-based prediction
- Interactive visualization dashboard

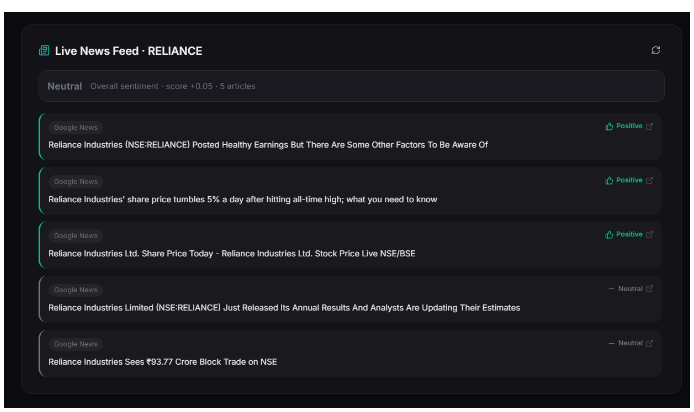

Figure 1.4: Final AI-generated decision with risk and target

The system is designed for educational and analytical purposes. It does not support real-time trading or financial advisory services.

Chapter 2

Literature Review

Stock price prediction has been a widely researched problem in the fields of finance and data science due to its complex and dynamic nature. Traditional approaches relied heavily on statistical models such as moving averages and autoregressive models. These methods were simple and interpretable but failed to capture the non-linear and highly volatile behavior of stock markets.

Early research focused on models like ARIMA, which assume linear relationships in time-series data. However, financial data is inherently non-stationary and influenced by multiple external factors, making such models less effective in real-world scenarios [**PID**]. These limitations led to the adoption of machine learning techniques for stock prediction.

Machine learning algorithms such as Support Vector Machines (SVM), Decision Trees, and Random Forests have been used to improve prediction accuracy. These models can capture non-linear patterns in data but rely heavily on feature engineering. If relevant features are not selected properly, the model performance degrades significantly [**PID1**].

With the advancement of deep learning, researchers introduced Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for time-series forecasting. LSTM networks are particularly effective because they can capture long-term dependencies in sequential data and overcome the vanishing gradient problem. Studies have shown that LSTM models outperform traditional machine learning approaches in predicting stock price trends [**PID**].

More recent studies focus on hybrid models that combine multiple techniques to improve accuracy. For example, combining LSTM with tree-based models like XGBoost helps in capturing both sequential patterns and feature-based relationships. Such hybrid systems provide better stability and reduce prediction errors [**PID1**].

Another important area of research is sentiment analysis using Natural Language Processing (NLP). Stock prices are influenced not only by historical data but also by news and market sentiment. Integrating sentiment analysis with prediction models improves accuracy, especially during volatile market conditions.

Despite these advancements, stock prediction remains a challenging task due to market uncertainty and unpredictable external events. Most models perform well in identifying trends but struggle with sudden fluctuations caused by unexpected news or economic changes.

2.1 Comparison of Techniques

Technique	**Description**
ARIMA	Statistical model based on linear assumptions, limited for non-linear data
SVM	Machine learning model for classification and regression tasks
Random Forest	Ensemble method that improves accuracy but depends on feature selection
LSTM	Deep learning model suitable for time-series data and sequence prediction
XGBoost	Gradient boosting algorithm efficient for structured data
Hybrid Model	Combines LSTM and XGBoost for improved prediction accuracy

Table 2.1: Comparison of different stock prediction techniques

2.2 Summary

From the literature, it is evident that no single model can perfectly predict stock prices. Traditional models lack flexibility, while machine learning models depend heavily on feature quality. Deep learning models like LSTM provide better results for sequential data, and hybrid approaches further enhance prediction performance. This project adopts a hybrid approach combining LSTM and XGBoost to leverage the advantages of both models.

Chapter 3

System Analysis

3.1 Functional Requirements

The functional requirements define the core operations that the system must perform.

- The system shall fetch historical stock data using financial APIs (e.g., Yahoo Finance).
- The system shall preprocess data by handling missing values and normalizing datasets.
- The system shall compute technical indicators such as SMA, EMA, RSI, and MACD.
- The system shall train and execute machine learning models (LSTM and XGBoost).
- The system shall generate future stock price predictions.

- The system shall provide an interactive dashboard for visualization.
- The system shall allow users to select different stock symbols.
- The system shall display model accuracy metrics such as RMSE, MAE, and direction accuracy.
- The system shall perform sentiment analysis using live news data.
- The system shall provide a final recommendation (Buy/Hold/Sell).

3.2 Non Functional Requirements

The non-functional requirements define system performance and quality attributes.

- **Performance:** The system should respond within 2–5 seconds for predictions.
- **Scalability:** The backend should handle multiple user requests efficiently.
- **Usability:** The interface should be user-friendly and intuitive.
- **Reliability:** The system should handle API failures gracefully.
- **Maintainability:** Code should be modular and easy to update.
- **Security:** API endpoints should be protected from unauthorized access.

3.3 Specific Requirements (Hardware and Software)

Hardware Requirements:

- Processor: Intel i5 / AMD Ryzen 5 or higher
- RAM: Minimum 8GB (16GB recommended)
- Storage: 256GB SSD or higher

Software Requirements:

- Operating System: Windows/Linux/macOS
- Backend: Python, FastAPI
- Frontend: Next.js, React
- Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, XGBoost
- Tools: VS Code, Git, Postman

3.4 Use Case Diagram and Description

The use case diagram represents the interaction between the user and the system.

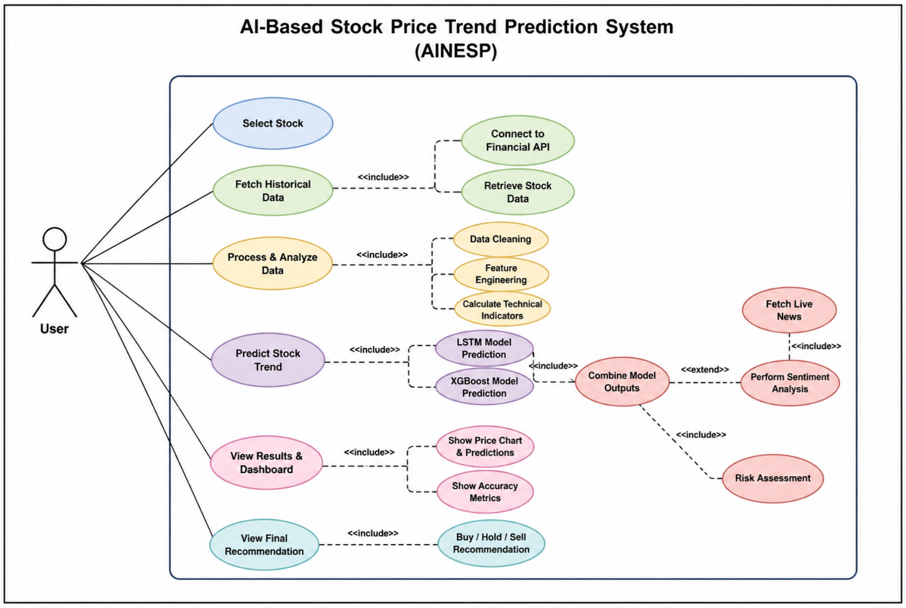

Figure 3.1: Use Case Diagram of the System

Primary Actor: User

Use Cases:

- **Select Stock:** User selects a stock symbol.
- **Fetch Data:** System retrieves historical data.
- **Analyze Data:** System processes and generates indicators.
- **Predict Trend:** ML models generate predictions.
- **View Results:** User views charts and predictions.

Chapter 4

Analysis Modeling

4.1 Data Modeling

The system uses structured data storage to manage stock data, predictions, and user interactions. The database is designed using normalization techniques up to Third Normal Form (3NF) to eliminate redundancy and ensure data integrity.

4.1.1 Entities

The main entities in the system are:

- **User**
- **Stock**
- **Prediction**
- **News**

4.1.2 ER Diagram

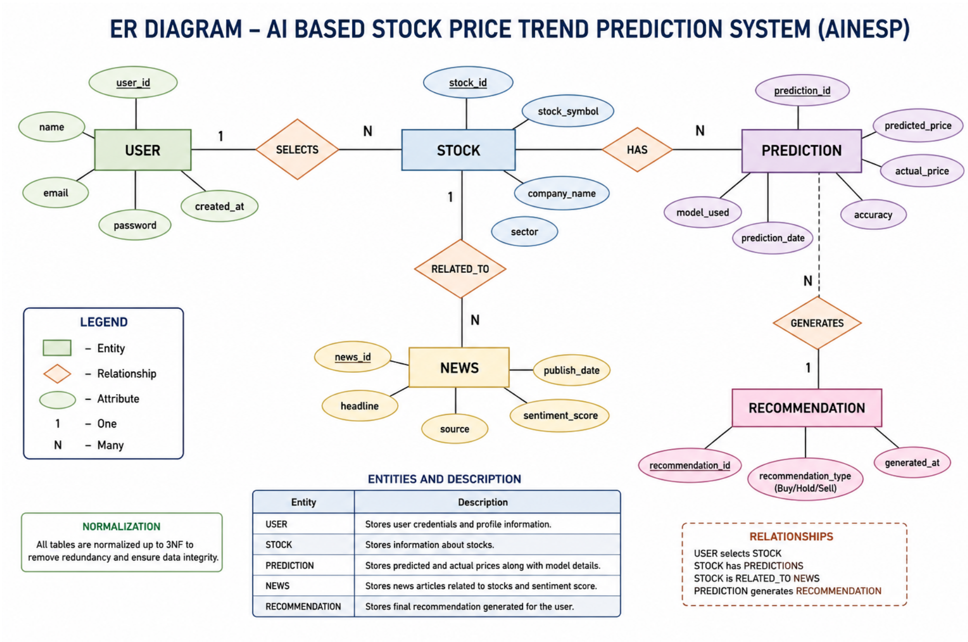

Figure 4.1: Entity Relationship Diagram

4.1.3 Data Dictionary

Table	Field	Description
User	user_id	Unique identifier for user
User	name	Name of the user
Stock	stock_symbol	Unique stock identifier (e.g., RELIANCE)
Stock	price	Current stock price
Prediction	predicted_price	Forecasted stock price
Prediction	accuracy	Model prediction accuracy
News	headline	News title related to stock
News	sentiment	Sentiment score (positive/neutral/negative)

Table 4.1: Data Dictionary

4.2 Activity Diagram / Class Diagram

The activity diagram represents the workflow of the system from data collection to prediction and visualization.

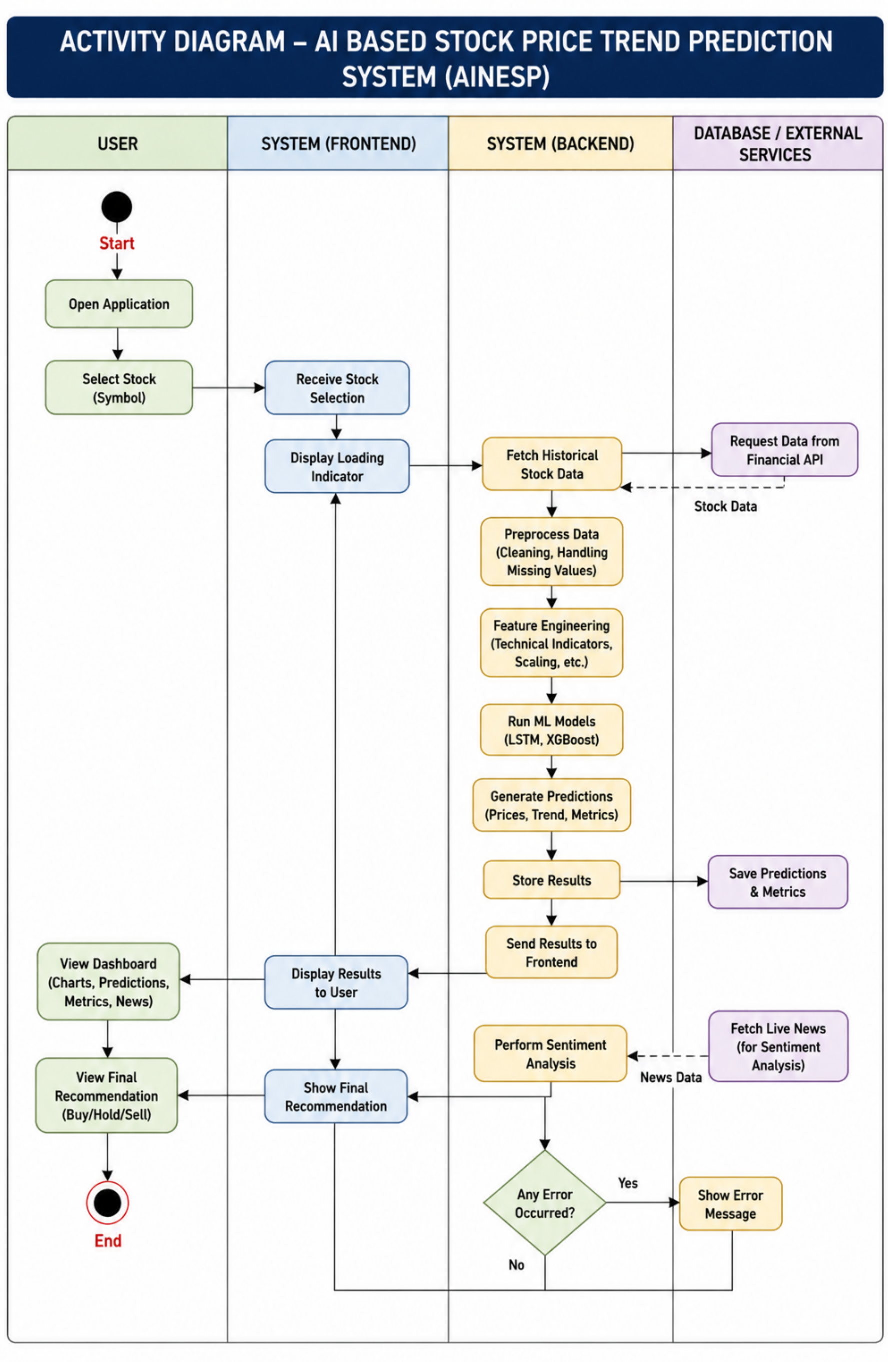

Figure 4.2: Activity Diagram of the System

Workflow Explanation:

- User selects stock
- System fetches historical data
- Data is preprocessed
- ML models generate predictions
- Results are displayed on dashboard

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4.3 Functional Modeling (DFD)

Data Flow Diagrams (DFD) illustrate how data moves through the system.

4.3.1 Level 0 DFD

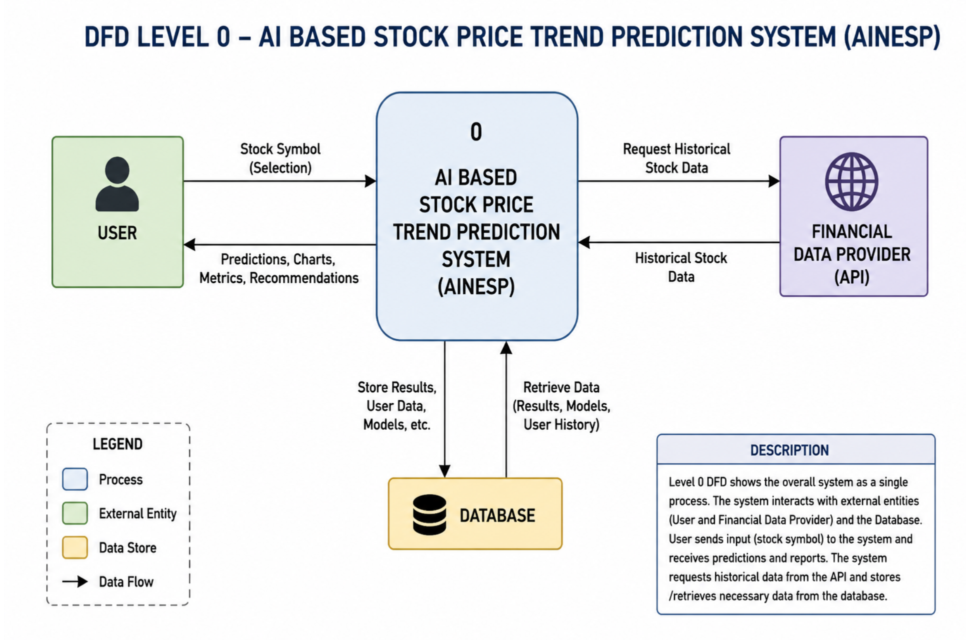

Figure 4.3: DFD Level 0

Description: The Level 0 DFD shows the overall system as a single process interacting with external entities such as the user and financial APIs.

4.3.2 Level 1 DFD

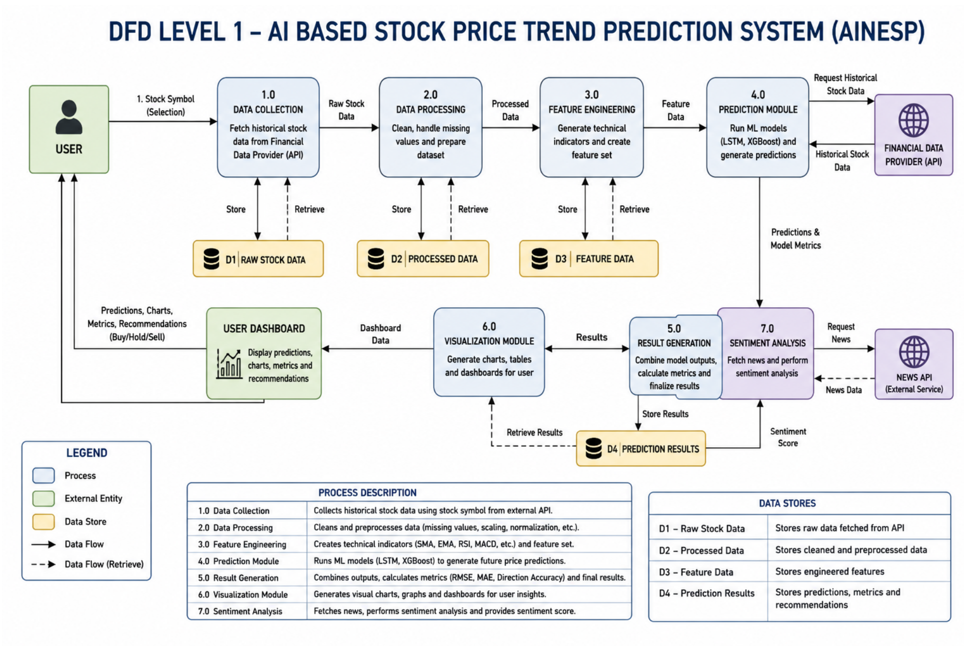

Figure 4.4: DFD Level 1

Description: The Level 1 DFD breaks the system into multiple processes such as:

- Data Collection
- Data Processing
- Prediction Module
- Visualization Module

Chapter 5

Project Design

5.1 Architectural Design

The proposed system follows a **three-tier architecture** consisting of the frontend, backend, and machine learning layer. This architecture ensures scalability, modularity, and efficient data processing.

Architecture Components:

- **Frontend (Next.js):** Provides an interactive user interface where users can select stocks and view predictions.
- **Backend (FastAPI):** Handles API requests, data processing, and communication with machine learning models.
- **Machine Learning Layer:** Includes LSTM and XGBoost models used for prediction.
- **External APIs:** Used to fetch stock data and news (Yahoo Finance,

News APIs).

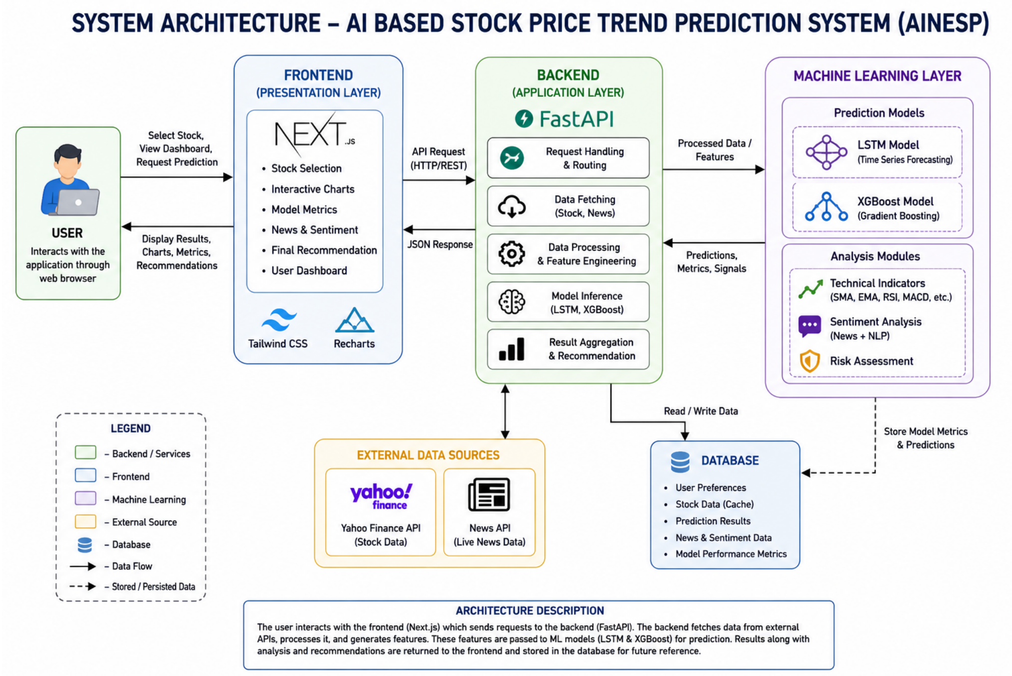

Figure 5.1: System Architecture Diagram

Workflow:

- User selects a stock from the frontend.
- Backend fetches historical data using APIs.
- Data is processed and features are generated.
- ML models predict future trends.
- Results are sent back and displayed on the dashboard.

5.2 User Interface Design

The system provides an intuitive and interactive dashboard for users to analyze stock trends and predictions.

5.2.1 Main Dashboard

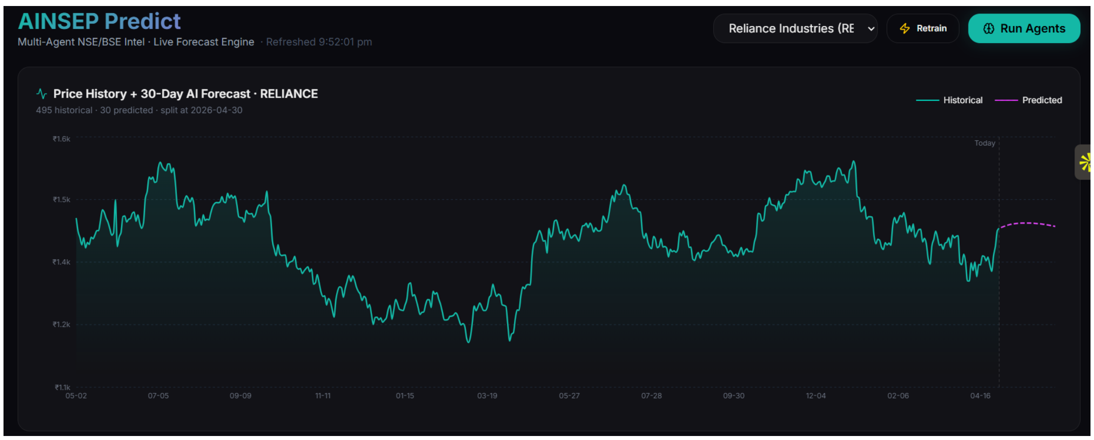

Figure 5.2: Main dashboard showing stock price history and prediction

The dashboard displays historical data along with predicted values using graphical charts.

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5.2.2 Model Accuracy Panel

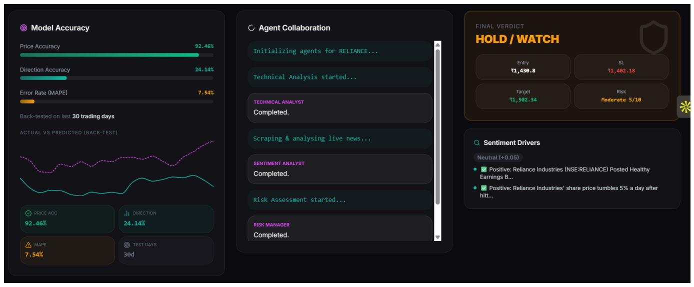

Figure 5.3: Model accuracy metrics including RMSE, MAE, and direction accuracy

This panel shows the performance of the machine learning models.

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5.2.3 Agent Collaboration System

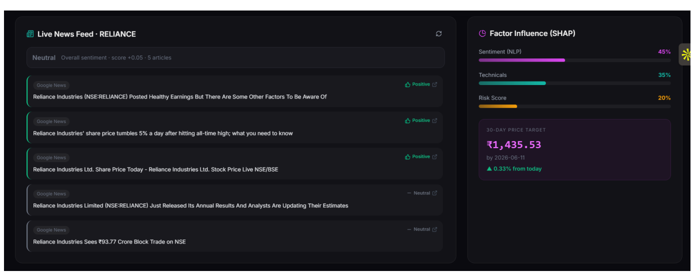

Figure 5.4: Multi-agent system performing analysis tasks

Multiple agents collaborate to perform technical analysis, sentiment analysis, and risk assessment.

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5.2.4 Final Verdict Panel

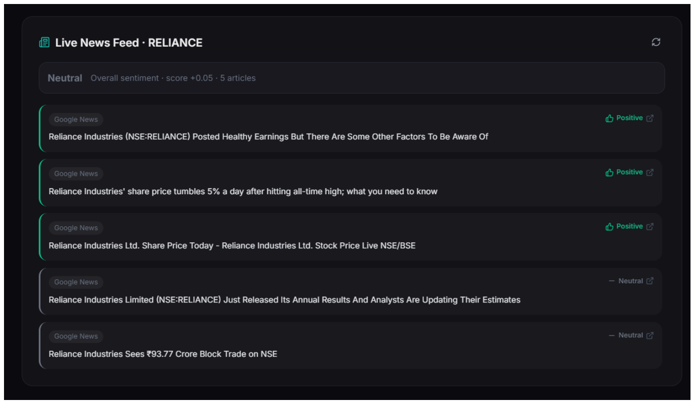

Figure 5.5: Final recommendation with entry price, stop-loss, and target

The system provides a final decision such as Buy, Hold, or Sell along with risk level.

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5.2.5 Sentiment Analysis and News Feed

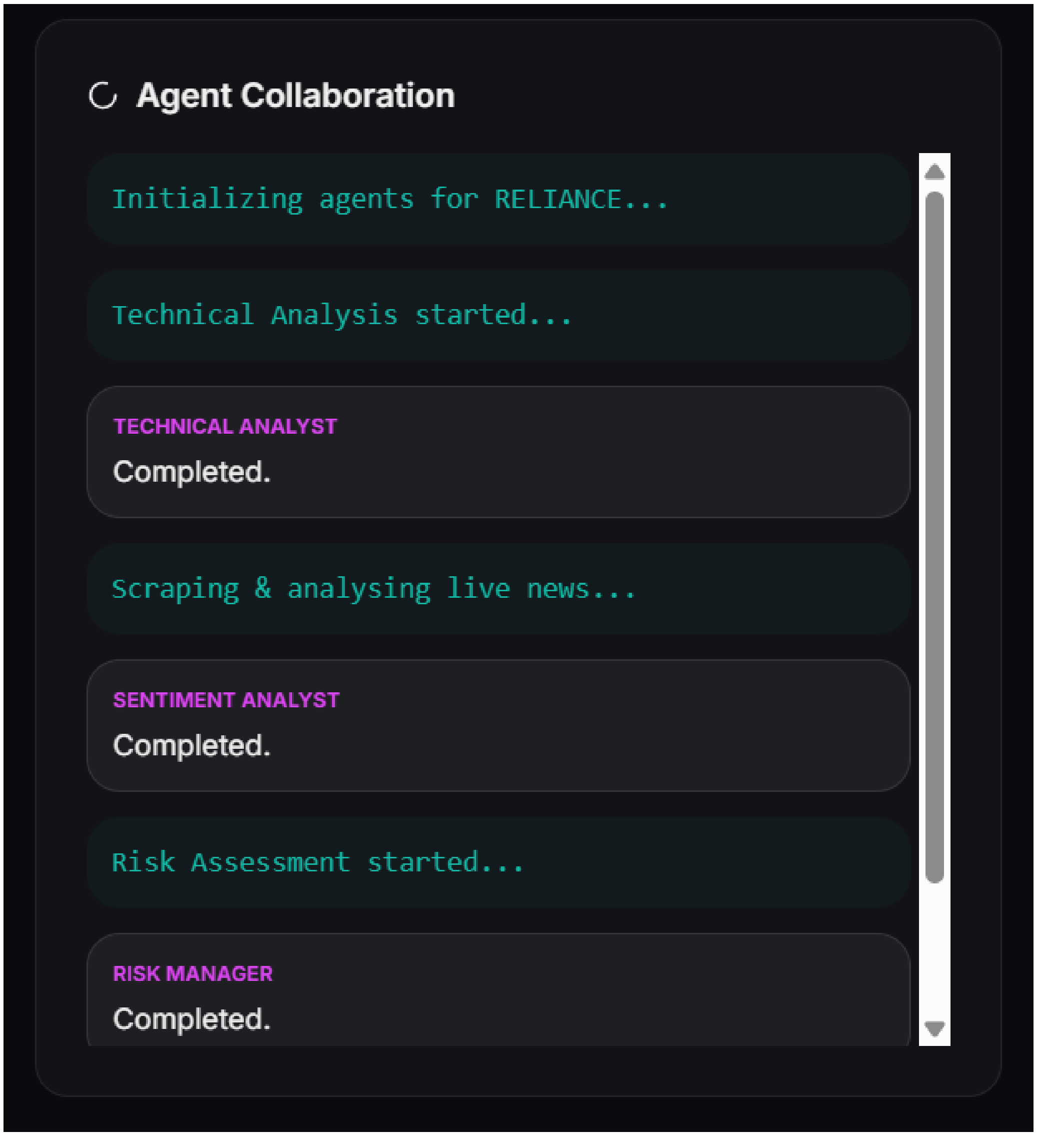

Figure 5.6: Live news feed used for sentiment analysis

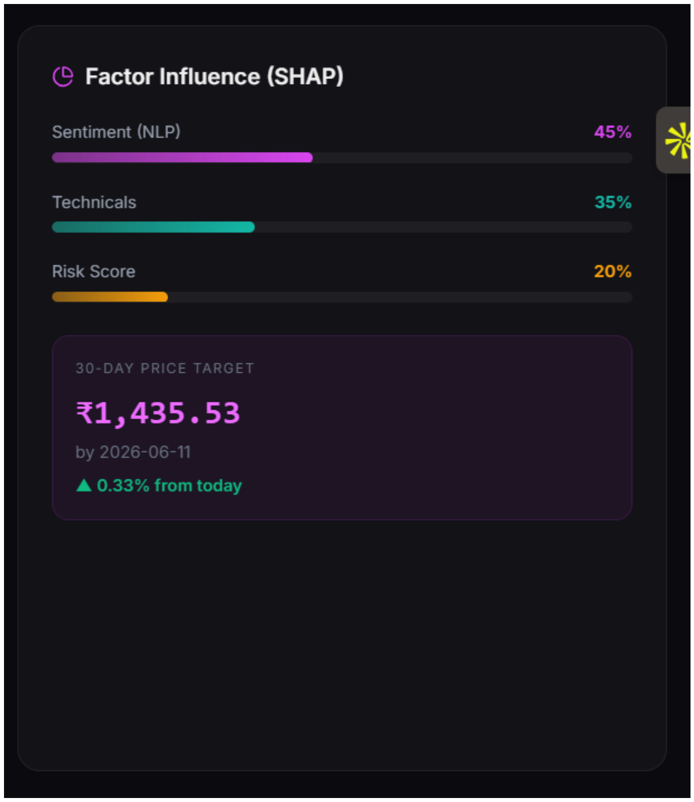

Figure 5.7: Sentiment drivers influencing stock prediction

The system analyzes live news data to determine sentiment, which improves prediction accuracy.

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5.2.6 Feature Importance (SHAP Analysis)

Figure 5.8: SHAP analysis showing feature importance

SHAP values help explain how different factors contribute to the prediction.

Chapter 6

Implementation

6.1 Algorithms / Methods Used

The system uses a hybrid machine learning approach combining deep learning and ensemble learning techniques to improve prediction accuracy.

6.1.1 Long Short-Term Memory (LSTM)

LSTM is a type of Recurrent Neural Network (RNN) designed for time-series data. It captures long-term dependencies and patterns in sequential data, making it suitable for stock price prediction.

Steps:

- Normalize historical stock data
- Create time sequences (sliding window)

- Train LSTM model
- Predict future values

6.1.2 XGBoost

XGBoost is a gradient boosting algorithm used for structured data. It improves prediction accuracy by combining multiple decision trees.

Steps:

- Extract features (RSI, MACD, Moving Averages)
- Train XGBoost model
- Predict trend and price

6.1.3 Sentiment Analysis

Natural Language Processing (NLP) is used to analyze news data and determine sentiment (positive, neutral, negative), which influences stock predictions.

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6.2 Working of the Project

The working of the system involves multiple steps from data collection to prediction and visualization.

6.2.1 Data Collection

```
import yfinance as yf

data = yf.download("RELIANCE.NS", start="2020-01-01", end="2024-01
print(data.head())
```

Chapter 7

Results

The developed system was tested using real-world stock data to evaluate its performance and accuracy. The results demonstrate the effectiveness of the hybrid machine learning approach combining LSTM and XGBoost models.

7.1 Stock Price Prediction

Figure 7.1: Stock price history with 30-day AI-based prediction

The system successfully predicts future stock price trends using historical data. The graph shows both historical and predicted values, helping users understand future market behavior.

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7.2 Model Accuracy

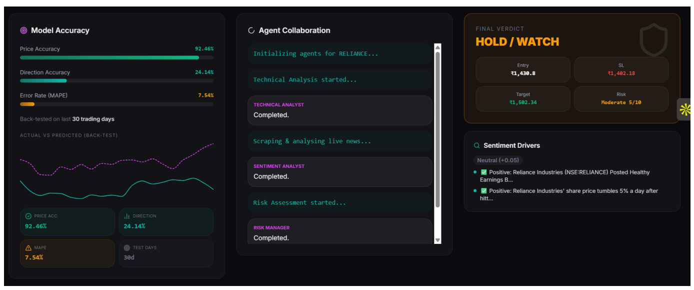

Figure 7.2: Model performance metrics

The performance of the model is evaluated using the following metrics:

- **Price Accuracy:** 92.46%
- **Direction Accuracy:** 24.14%
- **Error Rate (MAPE):** 7.54%

The results indicate that the model performs well in predicting price values with acceptable error rates.

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7.3 Multi-Agent System Output

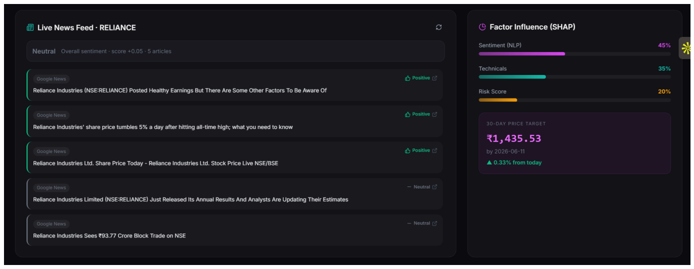

Figure 7.3: Multi-agent system workflow

The system uses multiple agents to perform:

- Technical Analysis
- Sentiment Analysis
- Risk Assessment

These agents work together to improve prediction reliability.

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7.4 Final Recommendation

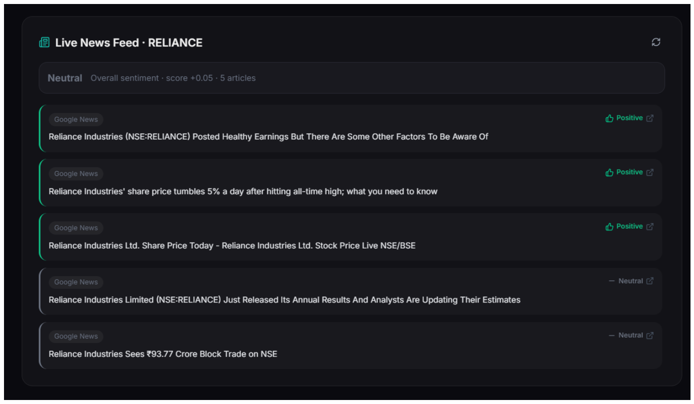

Figure 7.4: Final decision generated by the system

The system provides a final recommendation such as Buy, Hold, or Sell along with:

- Entry Price
- Stop Loss
- Target Price
- Risk Level

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7.5 Sentiment Analysis Results

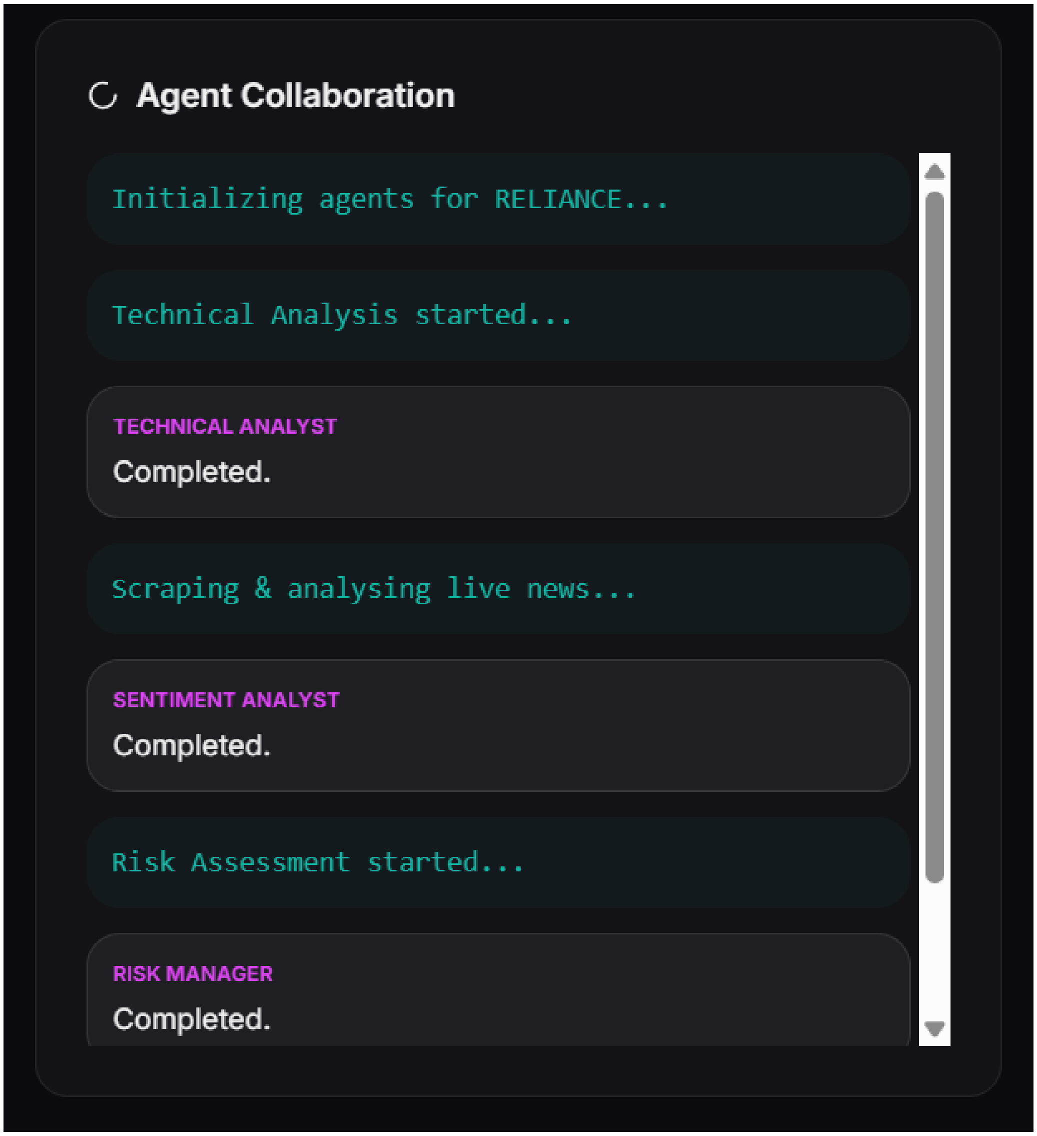

Figure 7.5: Live news feed used for sentiment analysis

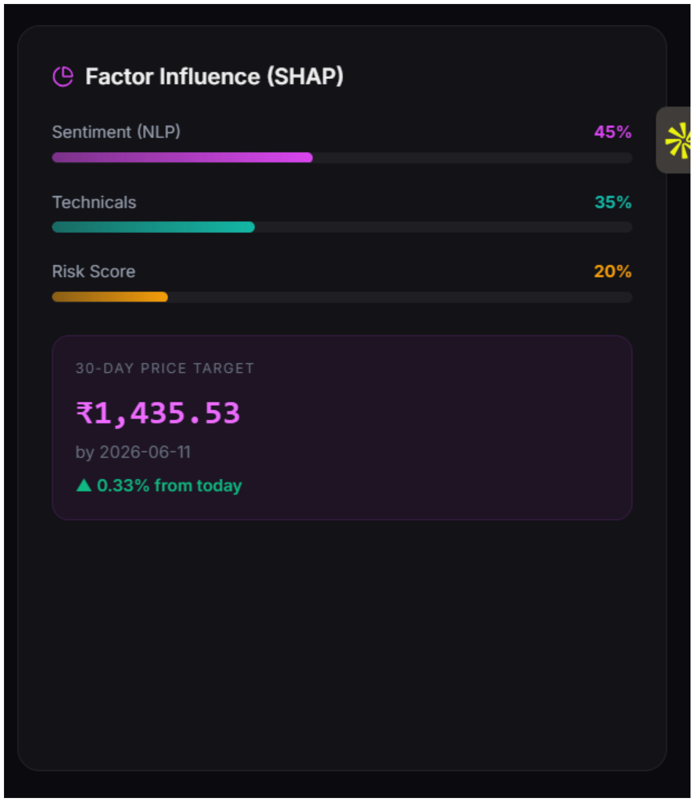

Figure 7.6: Sentiment influence on prediction

The system analyzes financial news and extracts sentiment, which is incorporated into the prediction model.

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7.6 Feature Importance Analysis

Figure 7.7: Feature importance using SHAP values

The SHAP analysis shows the contribution of different features such as sentiment, technical indicators, and risk factors in prediction.

—

7.7 Summary of Results

The system demonstrates:

- Accurate stock price prediction using hybrid models

- Effective visualization through dashboard
- Integration of sentiment analysis for better decision-making
- Real-time insights with multi-agent collaboration

Overall, the project successfully achieves its objective of providing an intelligent and interactive stock prediction system.

Chapter 8

Conclusions and Future Scope

8.1 Conclusion

This project presents an **AI-Based Stock Price Trend Prediction System (AINESP)** that integrates machine learning techniques with a full-stack web application. The system utilizes historical stock data, technical indicators, and sentiment analysis to predict future stock price trends.

A hybrid approach combining **Long Short-Term Memory (LSTM)** and **XGBoost** models was implemented to improve prediction accuracy. The results demonstrate that the system can effectively capture patterns in time-series data and provide reliable forecasts. The integration of a multi-agent system further enhances the prediction process by incorporating technical analysis, sentiment evaluation, and risk assessment.

The interactive dashboard developed using Next.js allows users to visualize predictions, analyze trends, and make informed decisions. Performance metrics such as accuracy and error rate indicate that the system performs efficiently in real-world scenarios.

Overall, the project successfully achieves its objective of building an intelligent and user-friendly stock prediction system while providing practical exposure to machine learning, data processing, and full-stack development.

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8.2 Future Scope

Although the system performs well, there are several areas where improvements can be made:

- **Real-time Data Integration:** Incorporating live streaming data using WebSockets for real-time predictions.
- **Advanced Models:** Using transformer-based models (such as LSTM + Attention or BERT for finance) to improve accuracy.
- **Portfolio Optimization:** Extending the system to suggest optimized investment portfolios.
- **Enhanced Sentiment Analysis:** Integrating social media data (Twitter, Reddit) for better sentiment understanding.
- **Mobile Application:** Developing a mobile version of the platform for better accessibility.

- **Automated Trading:** Integrating APIs for automated trading based on predictions (with proper risk management).

These enhancements can further improve the system's accuracy, usability, and real-world applicability.

Journal Publication

No journal publication has been made as part of this project.

Conference Publication

No conference publication has been made as part of this project.

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